Alina Du

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Education

Columbia University Expected May 2028
B.S. Computer Science; Minors in Financial Engineering, Math | GPA: 3.9/4.0 New York, NY

- **Coursework:** Machine Learning, Optimization, Software Quality and Design, Data Structures
- **Activities:** Teaching Assistant; Columbia Lioness Quantitative, WiCS, SWE

Awards & Skills

Awards: Jane Street WiSE (top 60); MIT Math Prize for Girls; 2x AIME Qualifier; 8x Waterloo math contest top 1%; Waterloo National Math Scholarship (15 recipients)
Languages: Python, Java, TypeScript, JavaScript, SQL
ML/Data: PyTorch, Scikit-learn, Hugging Face, NumPy, Pandas, SciPy
Frameworks: React/React Native, Next.js, FastAPI, LangGraph
Backend: AWS, Google Cloud, Vercel, PostgreSQL, MongoDB, Git

Experience

Data and ML Researcher Oct 2026 - Present
Columbia Plasma Physics Lab New York, NY

- Built a Codabench prototype for a CASP-style plasma prediction challenge on ∼15K DIII-D tokamak shots, with a custom Python scoring program and hidden answer keys
- Developing latent-space analysis of sparse data regions to predict out-of-distribution shots

Research Assistant – Data Engineering Sep 2026 - Present
Columbia University New York, NY

- Built the study UI and GPT-5 mini chat integration for the lab's econ-LLM experiments, designing 150 chat conditions across 3 study versions
- Built a Python pipeline matching 155 OpenRouter models to Epoch AI capability scores, joining benchmark results, pricing, and provider data with versioned provenance

Machine Learning Research Intern May 2026 – Sep 2026
Columbia University New York, NY

- Raised reliability from 91.8% to 96.5% on 50 reasoning trees with risk-targeted verification
- Benchmarked 8 LLMs on 14K+ claims; most stayed 66–95% confident on wrong answers
- Presented research poster at the Jane Street and Columbia Research Fair

Software Engineer Intern Mar 2026 - Jun 2026
Intelligence Cubed New York, NY

- Built invite-code access control for the deployment flow, enforced on both frontend and backend
- Wrote a Node.js proxy routing Anthropic API calls to Cloudflare-hosted Kimi K2 for model swaps
- Built a single backend API that lets users deploy AI models to AWS, GCP, and Cloudflare

Software Engineer Intern Aug 2025 - Mar 2026
MeloMed – Health Tech Startup New York, NY

- Built React Native interfaces delivering sentiment-driven recommendations to 1,200+ users
- Reduced infrastructure costs by 23% through serverless AWS deployment and caching
- Cut PostgreSQL query latency from ∼800 to ∼150 ms via indexing and query rewrites

Projects

Pac-Manhattan | JavaScript, Three.js, Python, Shapely, Gemini API, ElevenLabs

- Rendered 171K NYC buildings in 3D with Three.js and A* pathfinding over a 27K-edge street graph
- Wrote a Python/Shapely pipeline turning OpenStreetMap and aerial imagery into per-borough maps
- Added a Gemini pipeline that checks riddles for leaked answers, with ElevenLabs narration

Hack the Heart | Python/FastAPI, Next.js, OpenAI/Anthropic API, MongoDB, pytest

- Ran rule-based NLP (VADER, hedging) on full chats; an LLM reads the newest 10K messages
- Keyed date turns in MongoDB by (date ID, turn index) so failed dates resume without duplicate turns
- Imported chats from multiple sources with message-ID deduplication, keeping only the user's messages